

Experiment :-4

Problem statement:-

Implement error detection and correction mechanisms using block coding and Cyclic Redundancy Check (CRC). Simulate a communication system in Cisco Packet Tracer to demonstrate how errors are detected and corrected during data transmission.

Aim:- Error Detection and Correction Mechanisms Using Block Coding and CRC

Procedure:-

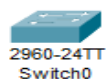
Step 1: Open Packet Tracer

- Launch Cisco Packet Tracer
- Go to **File > New**

Step 2: Add Devices to Workspace

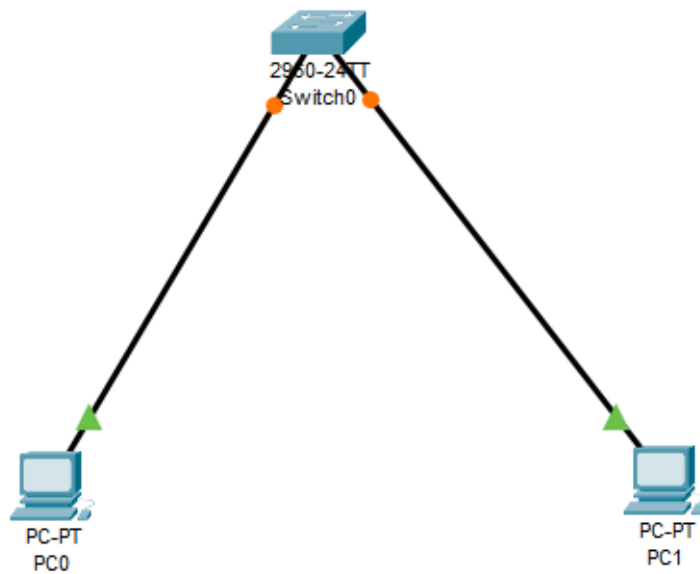
From the bottom device panel:

1. **Add 2 PCs** - Click "End Devices" category - drag 2 **PC icons** to workspace
2. **Add 1 Switch** - Click "Network Devices" - "Switches" - drag **2960 switch** to workspace

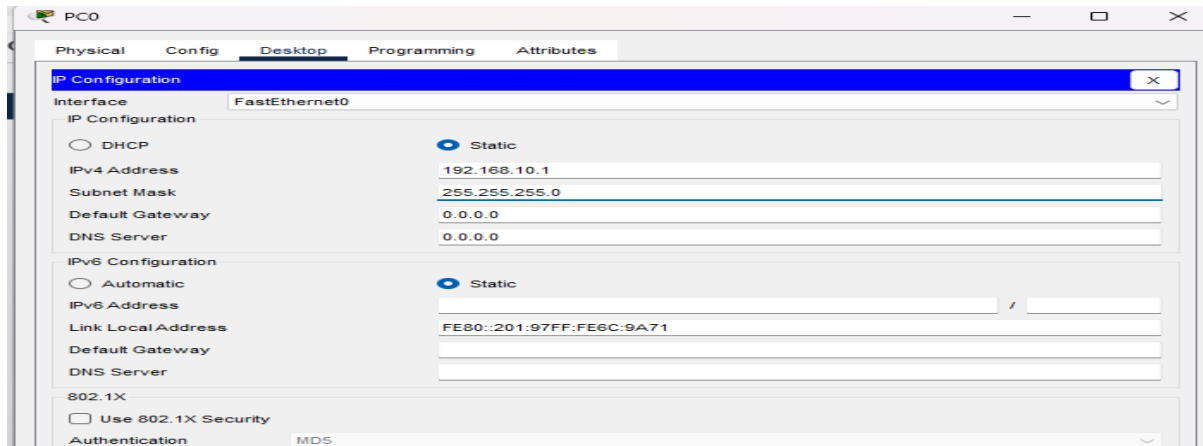


Step 3: Connect the Devices

- Select Connections tool (lightning bolt icon)
- Choose Copper Straight-Through cable
- Click PC0 - select FastEthernet0 port
- Click Switch - select FastEthernet0/1
- Repeat: Connect **PC1** - Switch **FastEthernet0/2**



Step 2: Configure IP Address



Step 7: Send a Simple PDU (Ping)

1. Click the **envelope icon** on the toolbar - "Add Simple PDU"
2. Click on **PC0** (source)
3. Click on **PC1** (destination)
4. You'll see a packet animation - this simulates data transmission
5. Check the **PDU List window** at the bottom — it should show "**Successful**"

